

Approximation Algorithms for Big Data

Introduction

Fall 2026-7

Welcome

- Class: Approximation Algorithms for Big Data

- Instructor: Tal Wagner

Today's Plan

- Introductions
- What is this class about?
- Administration
- Start Chapter 1: Metric spaces and dimension reduction

Introductions: Me

- Tal Wagner

- Contact:
  - Email: talwag@tauex.tau.ac.il
    - Please use this email for class-related correspondenc
    - Please don’t use Outlook auto-complete (I won’t get your email)
  - Class forum on Moodle (please use it if helpful to everyone)

Introductions: You



pe.app/talwagner

Introduction:

What this class is about

Algorithms in Intro to Algorithms

- Mostly:
  - Exact
  - Deterministic
  - “Efficient” = polynomial time (and space) in the input size n

Enter: Reality

- Data is huge
  - Time $O(n^2)$ is too slow
  - Even time $O(n)$ is often too slow
    - *What can we even do if we don't have time read the input??*



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    - *What can we even do if we don't have time read the input??*
  - Space $O(n)$ is too much
    - *We cannot even store the entire input*
- Data is high-dimensional
  - *"Curse of dimensionality"*



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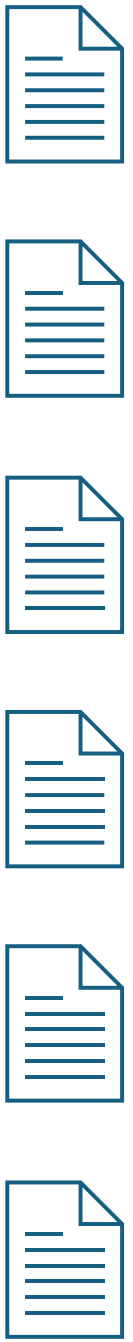
Time

- **Problem: Search**

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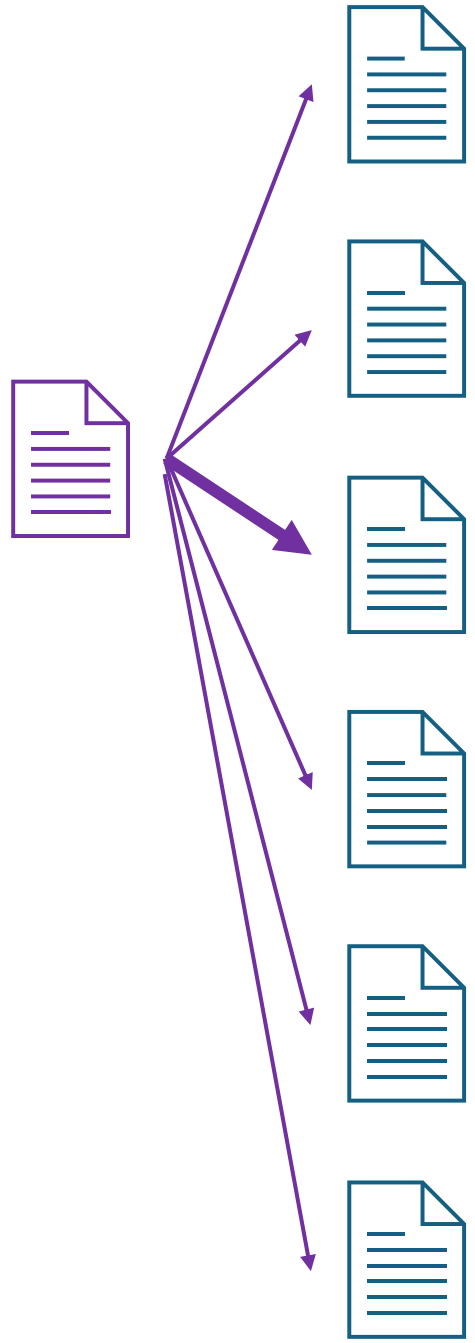
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  - Input: we have a big dataset of n objects



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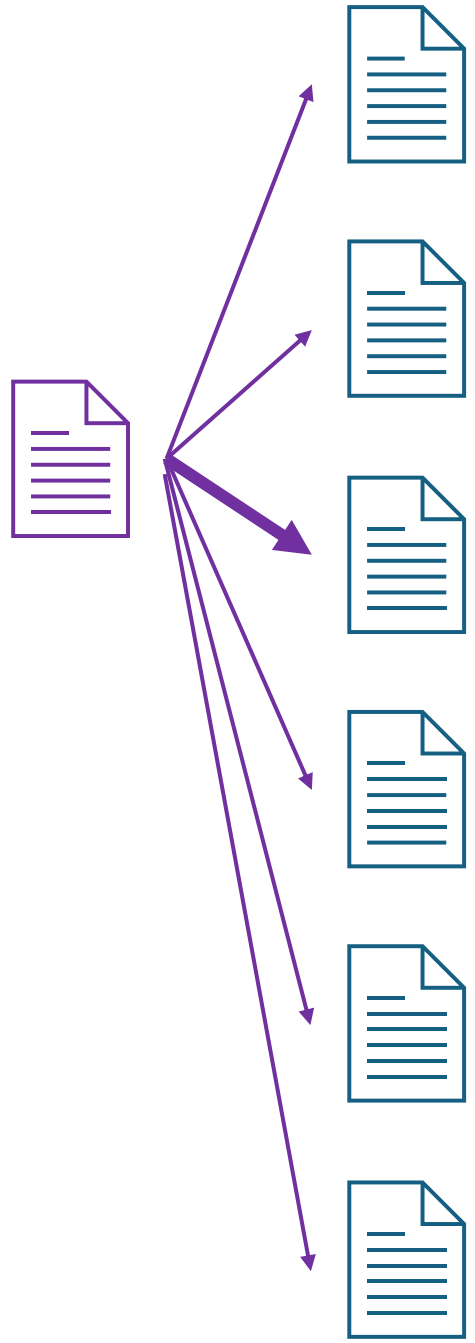
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  - Input: we have a big dataset of n objects
  - Goal: serve user queries
    - Get search query
    - Return best match from dataset



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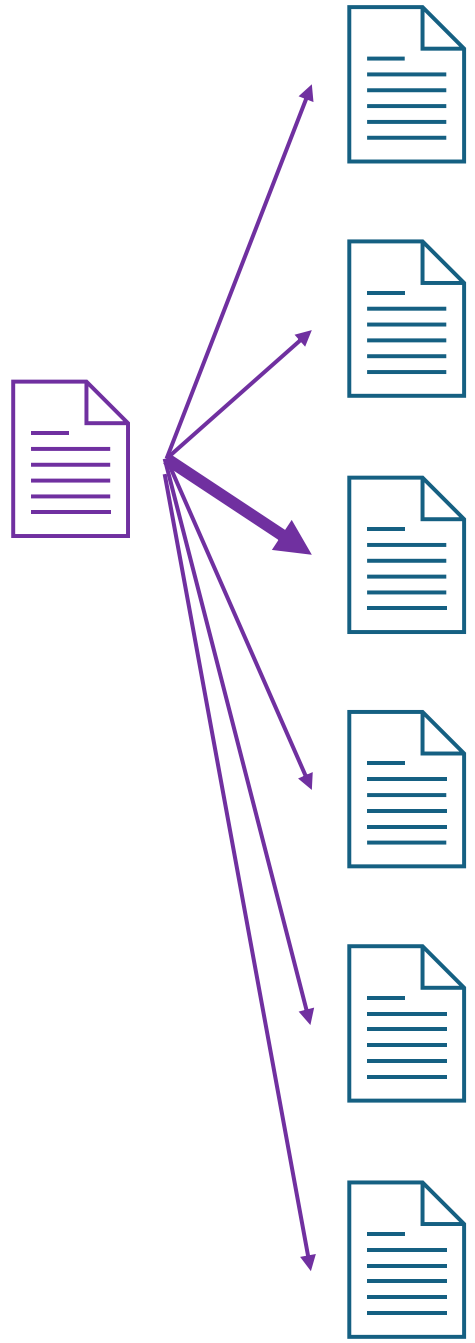
- Trivial solution?



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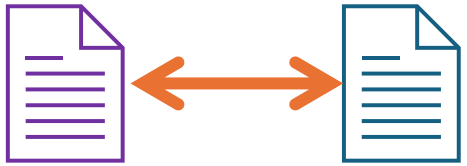
- Trivial solution?
  - Go over objects one by one per query
  - Time: $O(n)$ times pairwise match evaluation
  - **Is this efficient enough?**



Let’s see some examples:

Dimensionality

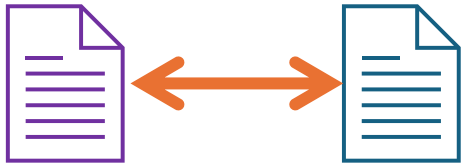
- What does it even mean that objects “match”?



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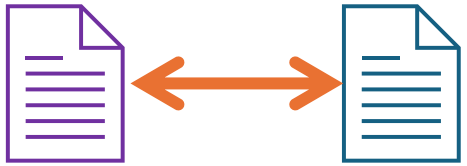
Dimensionality

- What does it even mean that objects “match”?
  - Texts on the same topic are “close”
  - Texts on unrelated topics are “far”
  - Images of the same objects are “close”
  - Images unrelated visually are “far”



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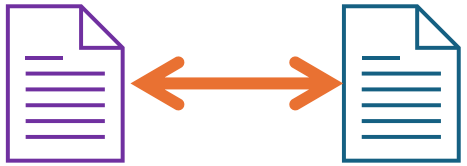


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- **Search** requires measuring **distance** between objects
  - Many problems we’ll see have a **metric structure**

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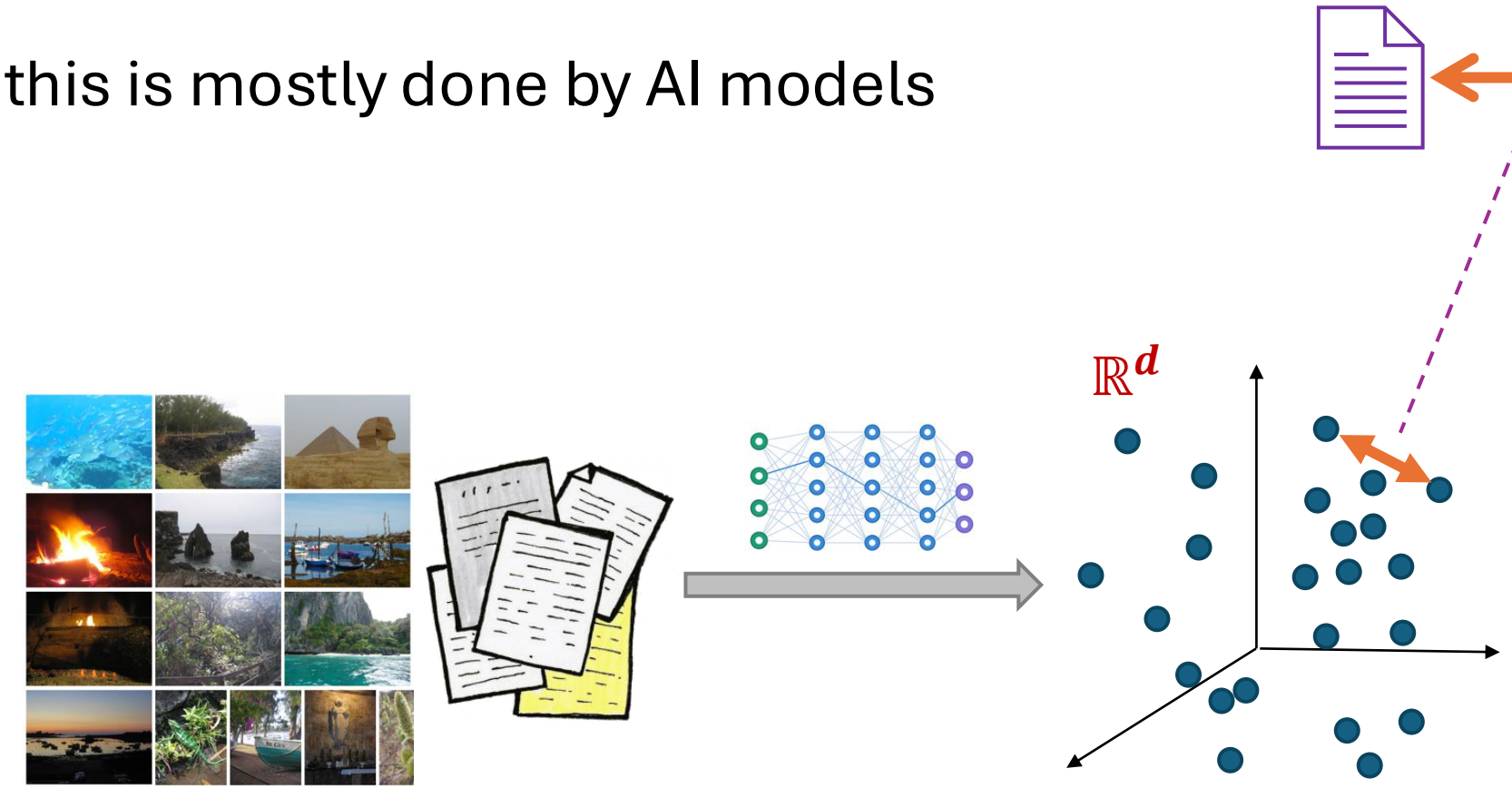
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  - Many problems we'll see have a **metric structure**
  - **Where does it come from?**

Let’s see some examples:

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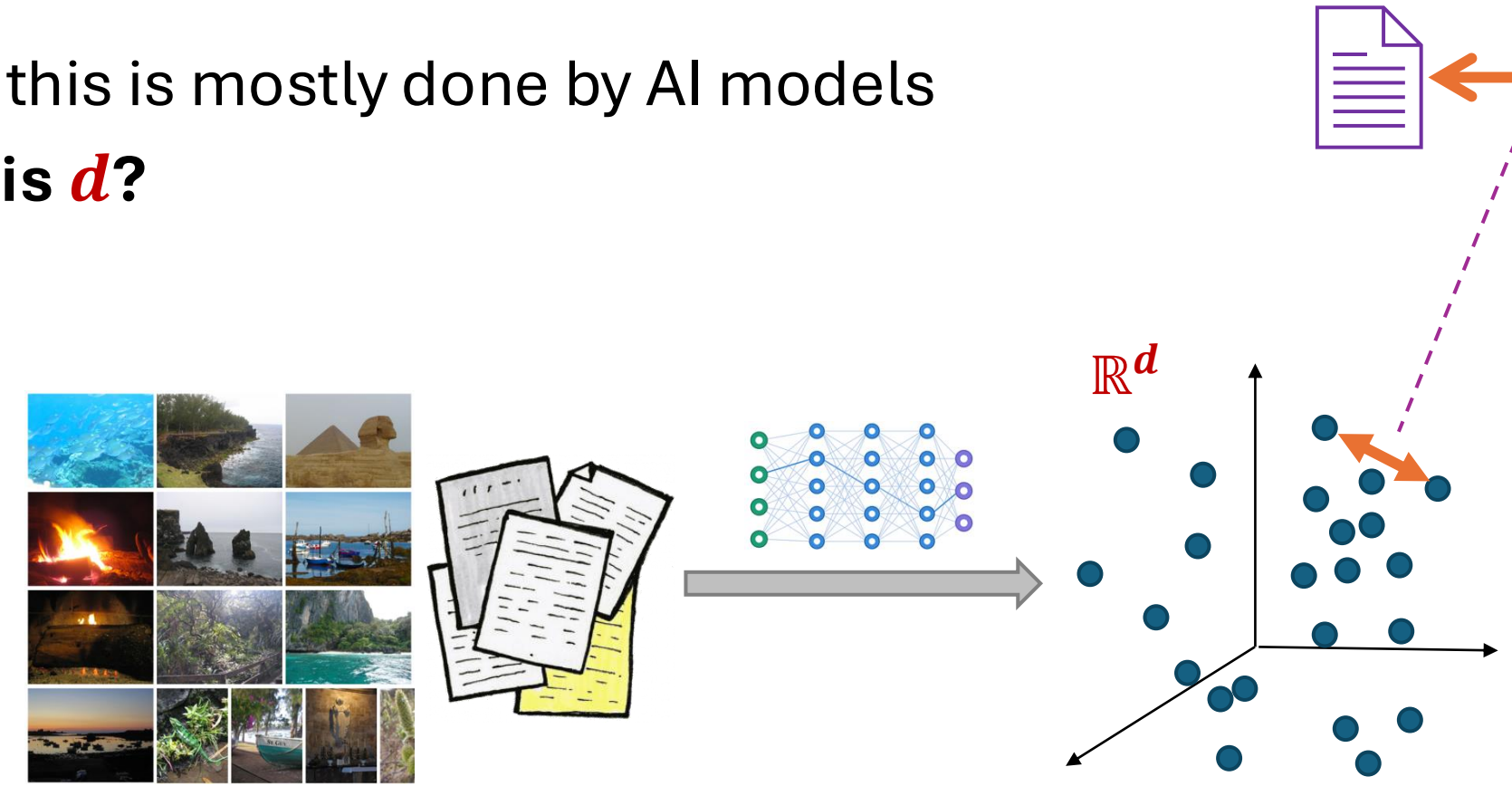
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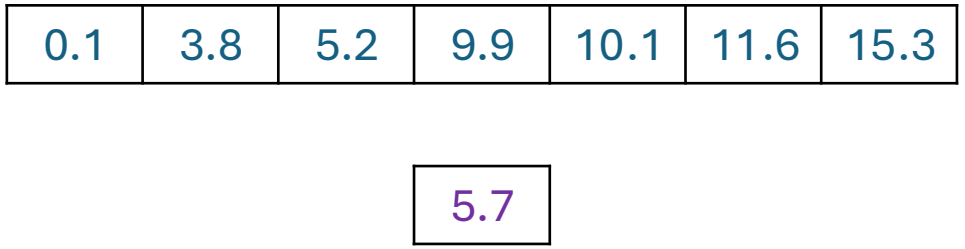
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- **What is d?**



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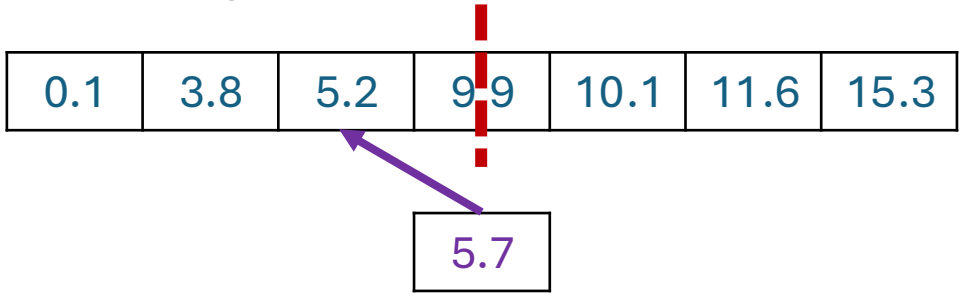
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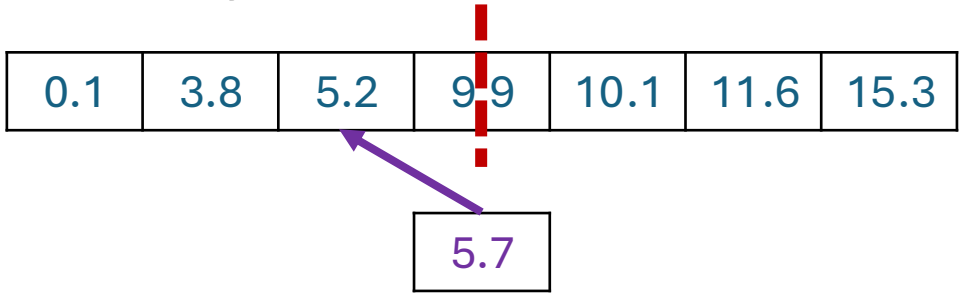
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  - Binary search:
    - In each step, split space into 2 parts
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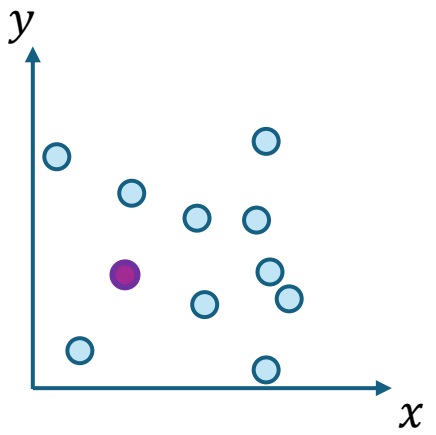
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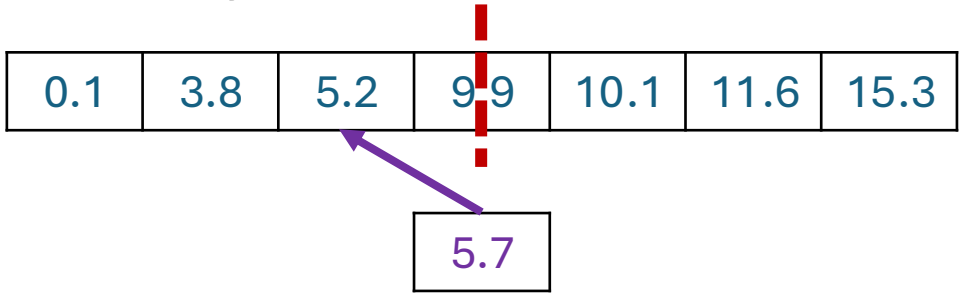
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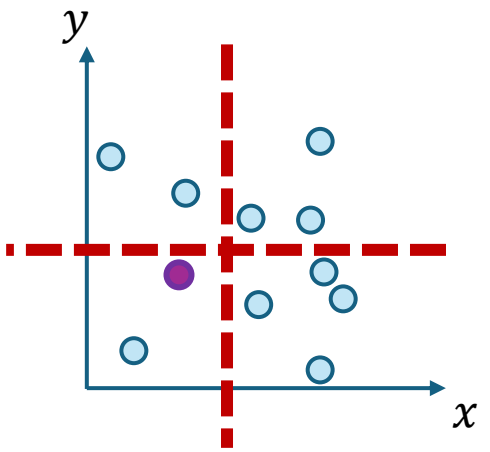
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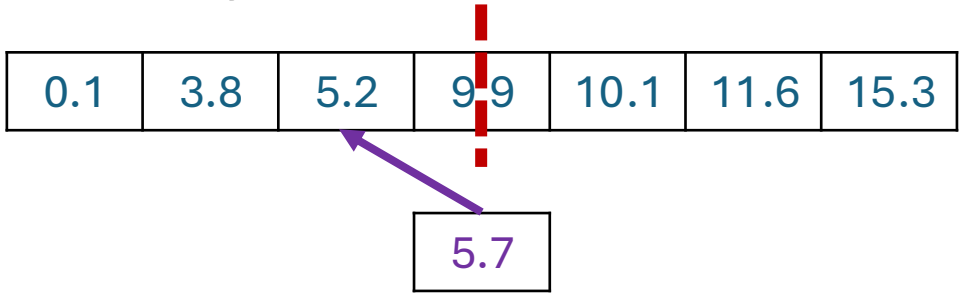
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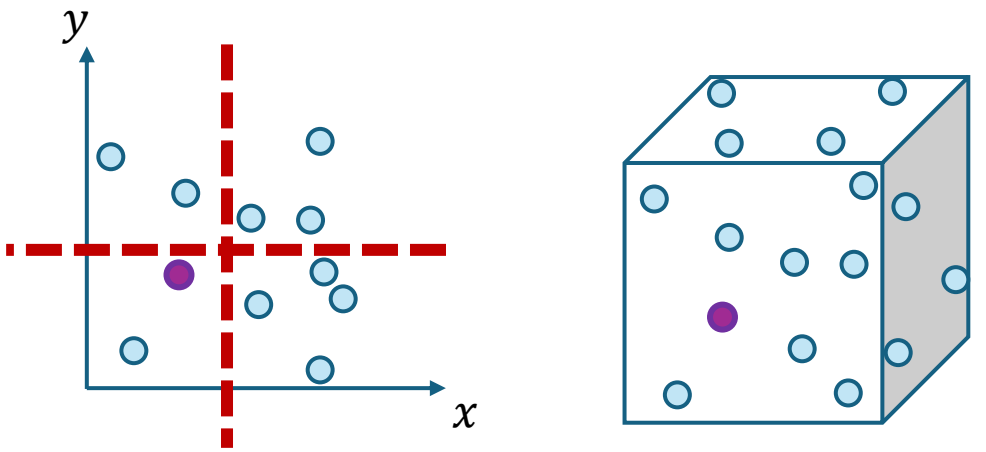
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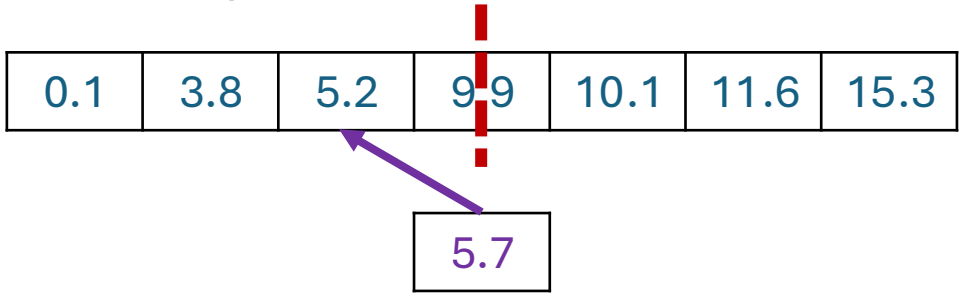
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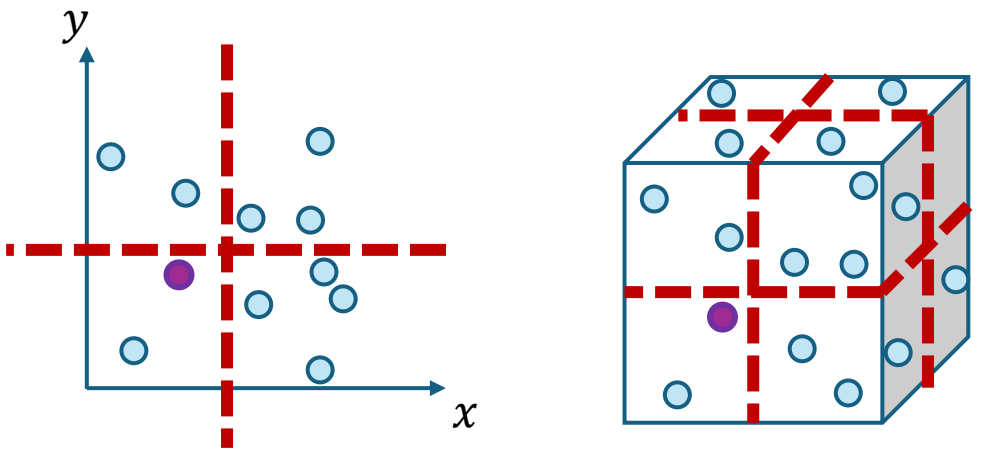
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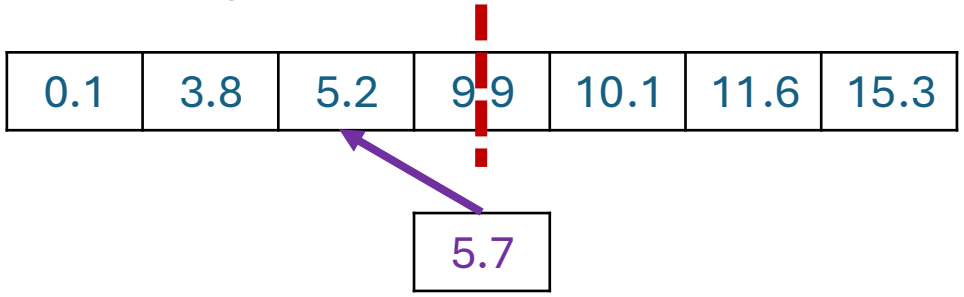
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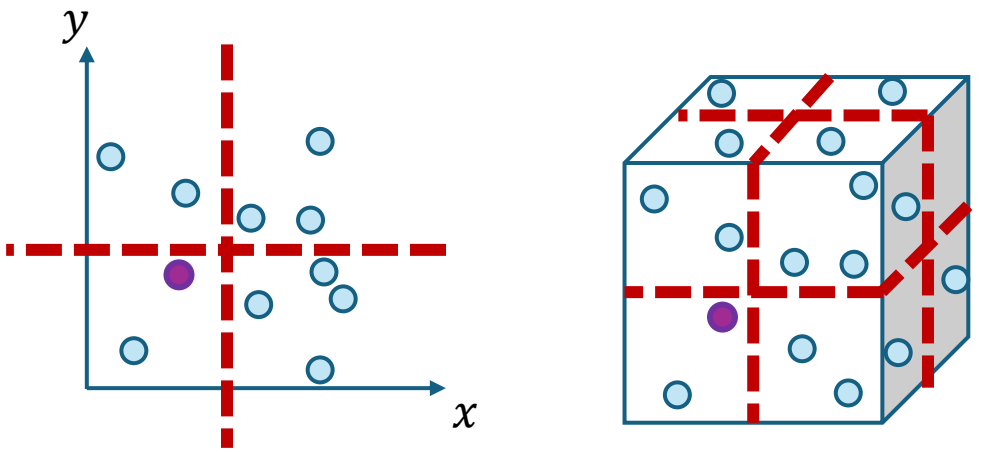
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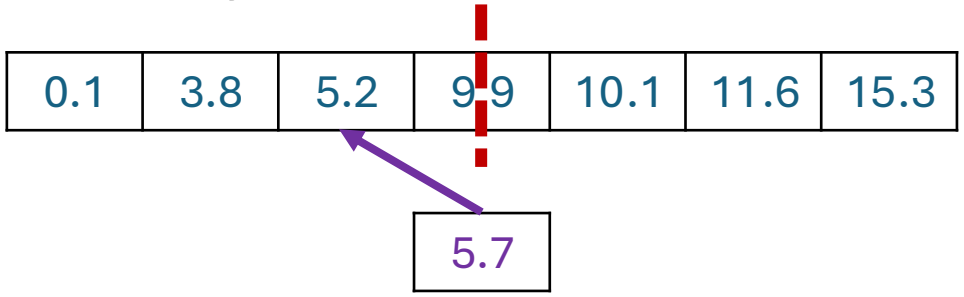
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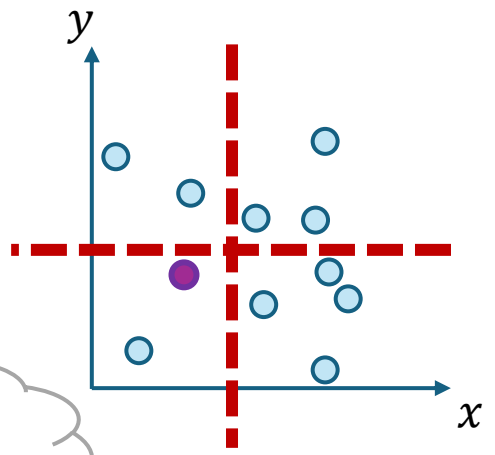
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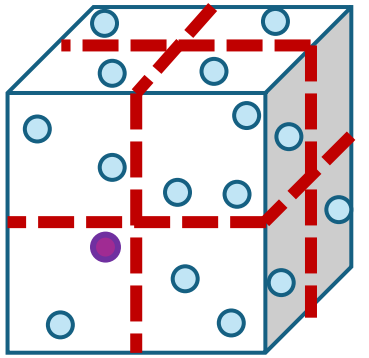
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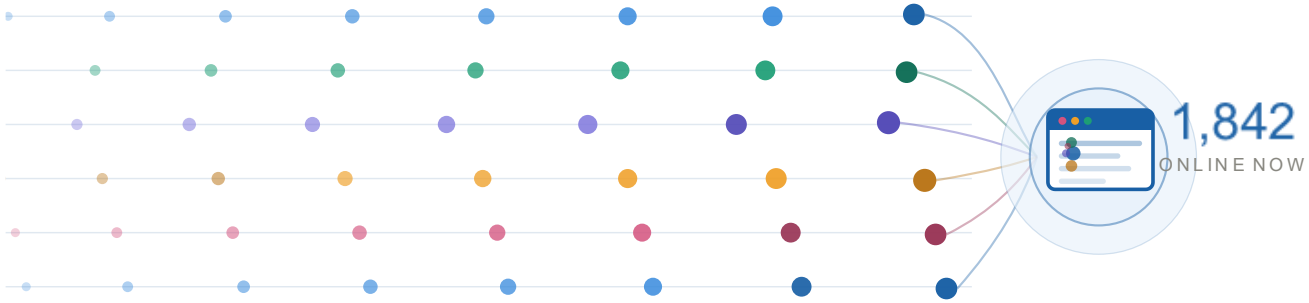
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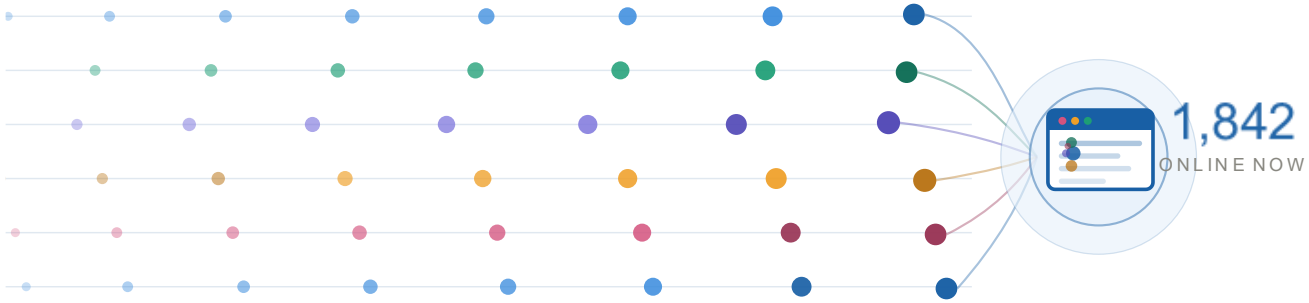
Curse of dimensionality

Let's see some examples:
Space



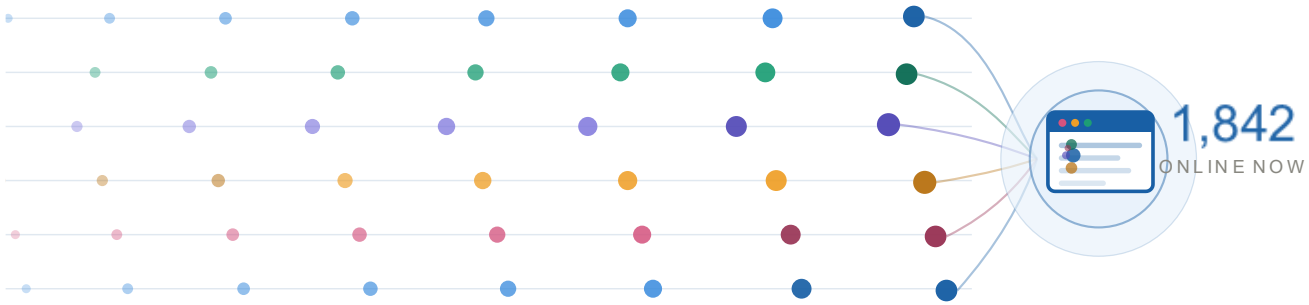
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  - Input: stream of n user visits to your web service
  - Goal: How many unique users visited today?

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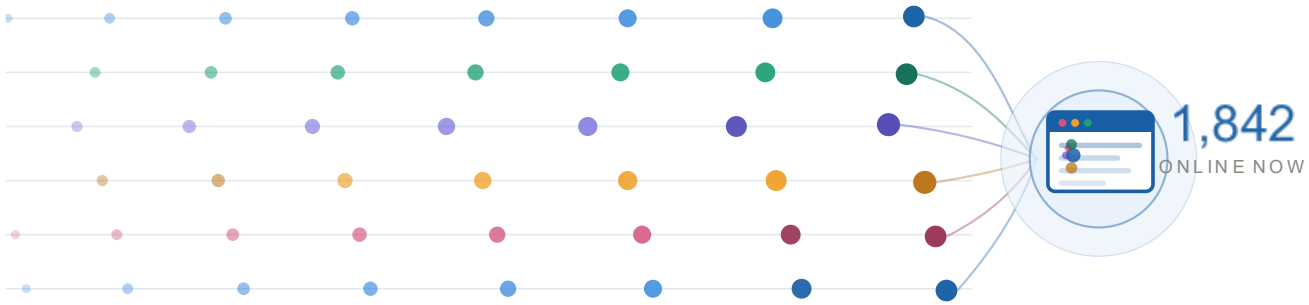
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    - **What if this is too much??**

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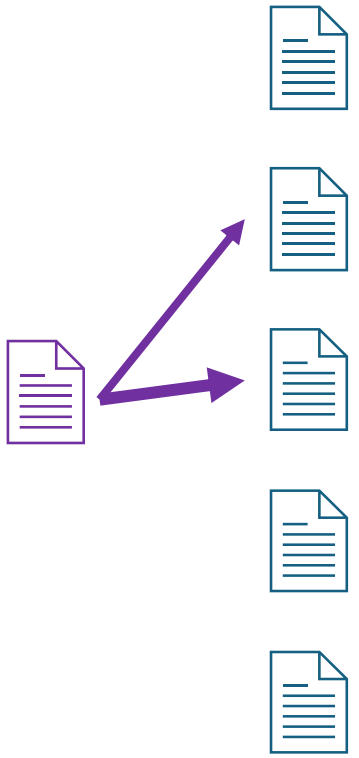
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- **Problem: Counting**
  - Input: stream of n user visits to your web service
  - Goal: How many users visited today?
  - Trivial solution: just count! Space: $O(\log n)$
    - **What if even this is too much??**

To the rescue:

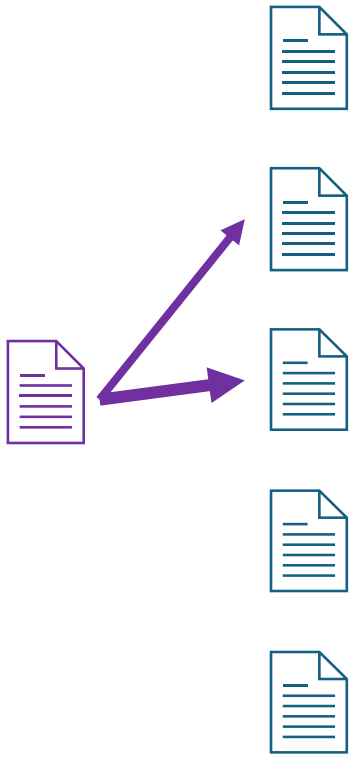
Approximation and Randomness





To the rescue:

Approximation and Randomness

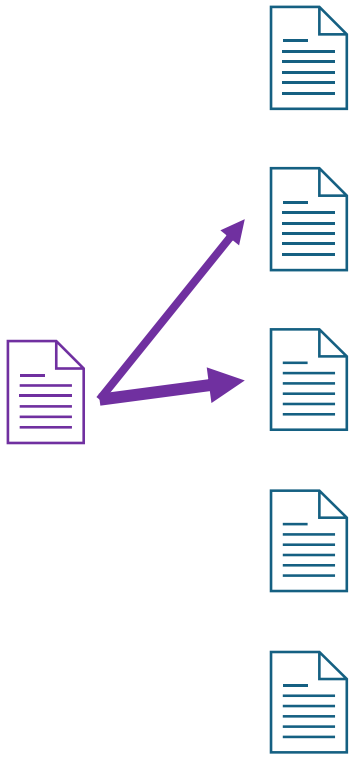


- **Approximation:**
  - **Search**: Do we really need the best match object?
    - Maybe other highly matching objects are good enough?



To the rescue:

Approximation and Randomness

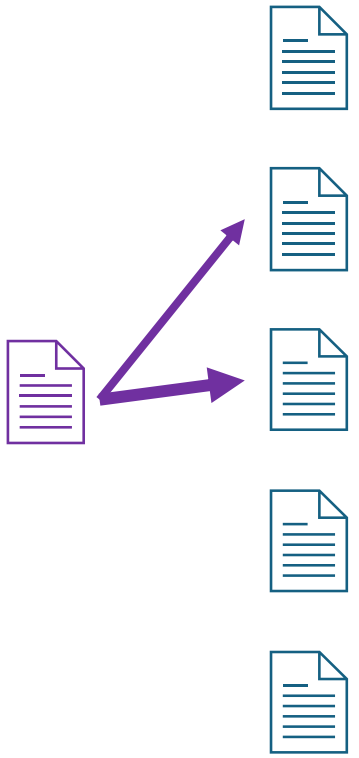


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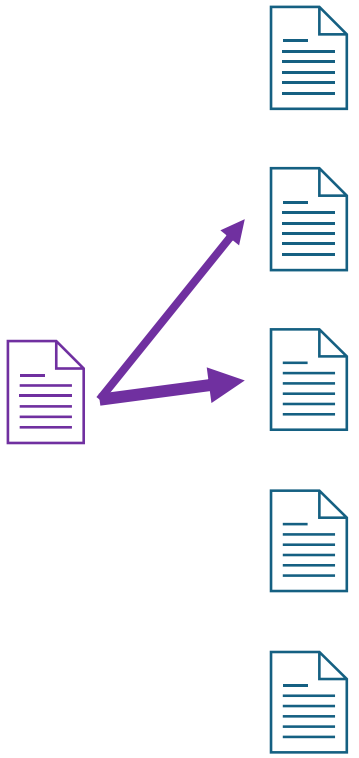


- **Approximation:**
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  - **Counting**: Do we really need the exact count?
  - Approximation is often sufficient
  - Enables dramatic efficiency improvements



To the rescue:

Approximation and Randomness



- **Approximation:**
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- **Randomness:**
  - Enables powerful algorithmic ideas
  - Chance of “bad” randomness can be made negligible

What this class is about:

- **The foundations of randomized approximation algorithms that scale to large and high-dimensional datasets**
  - **Theory:** *(the focus)*
    - Learn algorithmic methods
    - Formulate them mathematically and prove they work
  - **Practice:** *(we'll touch on it)*
    - Explore how they are implemented and how they behave on real data
    - Explore where theory and practice connect, and when they don't
  - **Conceptual goal:** Distill the **algorithmic idea** behind the math & code

List of Topics (subject to some change)

First "half" of the semester

- Metric embeddings, dimension reduction
- Sketching and streaming
- Nearest neighbor search
- Quadtree metrics

Second "half" of the semester

- Kernels in high dimension
- Efficient linear algebra

Introduction:

Administration

Class Goals

- Understand the key elements of the field
  - Key algorithmic methods and why they work
  - Computational models
  - Mathematical backdrop and proof techniques
  - Transition to implementation and how it looks in reality

- Dive deeper into a chosen topic, see what research in the area looks like
  - In final projects

Prerequisites

- The three big ones:
  - Algorithms and Data Structures (everything)
  - Linear algebra (norms, inner products, orthogonality, projections, eigenvectors and eigenvalues…)
  - Probability (all the basics; tail bounds: Markov, Chebyshev, …)
- Nice to have: any intro to ML
- The informal ones:
  - “Mathematical maturity” (comfort with formal proofs)
  - Scientific code (familiarity with NumPy / SciPy / PyTorch / etc.)
    - Or willingness to learn it

Setup

- Classes in person (no live zoom, but recordings uploaded)
  - Please ask questions, participate, engage
  - Please help me regulate volume and speed
- **Class website:** Moodle (slides, class forums, extra materials)
- No textbook, but Moodle links to lecture notes of overlapping classes
- This is an **advanced class**
- Grade:
  - 30% assignments (10% each)
  - 10% engagement (split into 4 parts)
  - 60% final project (half written report and half oral presentation)

Grade: Home Assignments (30%)

- Three assignments (equal weight)
- Each has mostly theory and a bit of implementation
- Submit in singles or group of 2-3

Grade: Engagement (10%)

- Rough idea:
  - Engage with the class actively and not just passively
  - Be part of class rather than a non-present observer
  - Make a small contribution to the “class community”
- You need to do this 4 times during the semester (equal grade weight):

1. Any topic other than the next three	3. Kernels
2. Nearest neighbor search	4. Efficient linear algebra

- How to fulfill?
  - Option 1: in-class attendance and participation during relevant lectures
  - Option 2: Record a short video explaining a concept (teaching aid)
  - Option 3: If you have other ideas I will hear you out and I may approve

Grade: Final Project (60%)

- Goal:
  - Dive deep into a topic from class
  - Engage with (relatively) recent research

- Project components:
  - Project proposal (binary approval, toward end of semester)
  - Written report (50% of project grade)
  - Oral presentation + interview (50% of project grade)

- Work in singles of in groups of 2-3

Grade: Final Project (60%)

- Project types:
  - **Deep reading:** explain paper in depth
    - The goal is not just to present a paper like in a seminar, but to do something beyond: explain the paper from a new angle, or explain how a few papers tie together, etc.
    - Will entail deep dive into mathematical proofs
  - **AI-assisted research:** make a scientific contribution to the field
    - Use AI to solve or make progress on an open problem in the area (now's the time!)
    - Present proof fully and formally in writing and in the presentation
    - Can fall back to a deep reading project if needed
    - Consider publication in some form is successful (with proper AI disclosure)
- Choose papers from provided list or on your own
- Reports and presentations will entail deep dive into mathematical proofs

Generative AI Usage

You may use generative AI tools in your homework assignments and final projects, under the following guiding principles:

Ownership

Accountability

Transparency

Generative AI Usage: Ownership

- Use AI tools to **support** your thinking, learning and writing – **not replace** them
- Maintain **ownership** of your work
- You should understand and be able to reproduce the solutions and results you submit
- Your submission in its totality should reflect **your own** intellectual work and understanding, not the AI's

Generative AI Usage: Ownership

Okay:

- Using AI to explain topics and concepts, and discuss papers and sources you've also engaged with directly
- Using AI for writing and coding assistance (emphasis on *assistance*)

Not okay:

- Submitting text/code that you not "own" intellectually, have full command of and can explain at a line-by-line level
- Relying on unverified AI-generated proofs, explanations or citations

Generative AI Usage: Accountability

- You are fully responsible for the content, accuracy, correctness and integrity of any work you submit – irrespective of any AI use
- You are accountable for any mistakes, inaccuracies, “hallucinated” references or results, etc
- Please know that signing one’s name on work one did not truly author is a severe academic transgression with potentially far-reaching consequences

Generative AI Usage: Transparency

- Include ”AI usage disclosures” in homework assignment or final projects

- In any case, a good rule of thumb:

Don’t use AI for anything you

wouldn’t want to admit you used AI for

Generative AI Usage: Informed Choices

- Ultimately, your AI use is your personal choice
- It is a great tool with great risks ("**cognitive debt**")

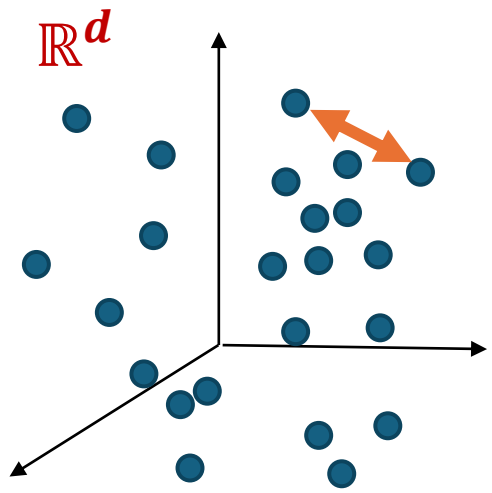
- You should weigh in each instance:
  - "Is it promoting my intellectual development or impeding it?"
  - "Am I gaining professional and scientific skills, or losing an opportunity to gain them?"
  - "Is this speed-to-depth trade off worth it?"
- The answer is often yes and often no – so ask yourself each time

Start of course material:

Metric Spaces and Dimensionality Reduction

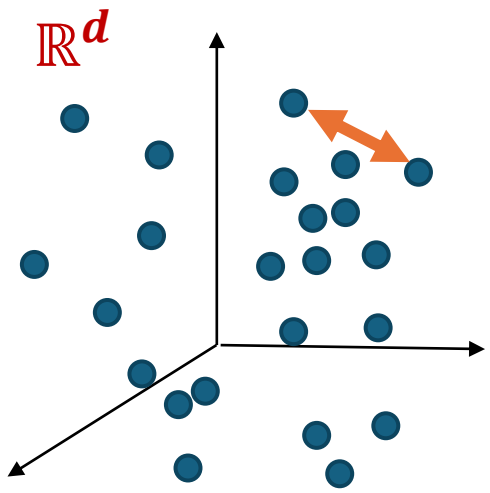
Metric spaces

- Definition: (X, D) is a **metric space** if
  - X is a set
  - D is a map $D: X \times X \to \mathbb{R}_{\geq 0}$
  - The following axioms are satisfied:
    - $D(x, x) = 0$ for every $x \in X$
    - $D(x, y) = D(y, x)$ for every $x, y \in X$
    - $D(x, y) \leq D(x, z) + D(z, y)$ for every $x, y, z \in X$



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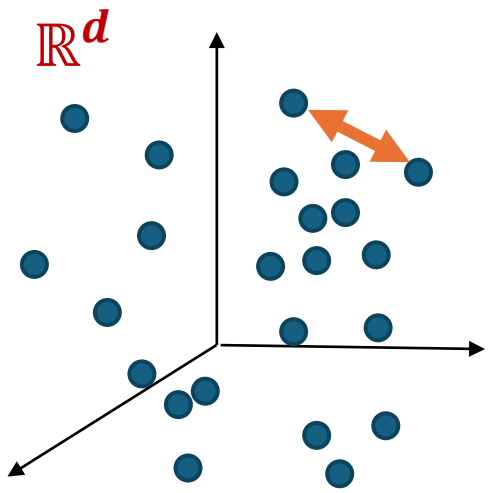
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- Norms induce metric spaces:
  - The ℓ_p-norm in $\mathbb{R}^d$: $\|x\|_p := (\sum_i |x_i|^p)^{1/p}$ for $p \in [1, \infty)$, and $\|x\|_\infty = \sup_i |x_i|$

Metric spaces

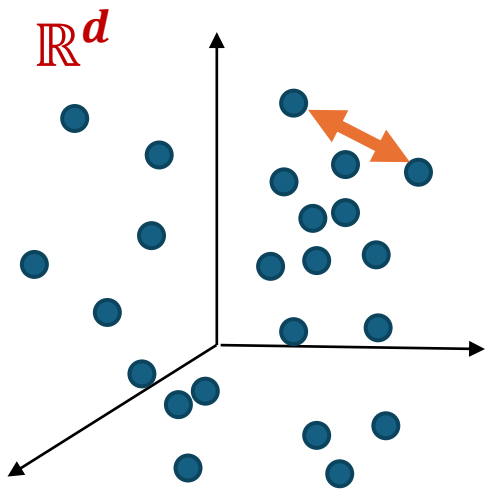
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  - Induced metric on $\mathbb{R}^d$: $D(x, y) = \|x - y\|_p$
  - **Euclidean** (ℓ_2): $\|x - y\|_2 = \sqrt{\sum_i |x_i|^2}$; **Manhattan** ($\ell_1$): $\|x - y\|_1 = \sum_i |x_i|$

Other Metrics

- Which other metrics or metric spaces have you seen?

Other Metrics

- Which other metrics or metric spaces have you seen?
  - Shortest path metric on graphs
  - Edit distance between strings
  - Total Variation distance between probability distributions
  - Wasserstein distance between distributions over metric spaces
  - …

Computing the Diameter

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- Trivial algorithm: Compute all distances and take the max
  - Cost: $\binom{n}{2} = O(n^2)$ distance computations
  - **Can we do better?**

Diameter Lower Bound

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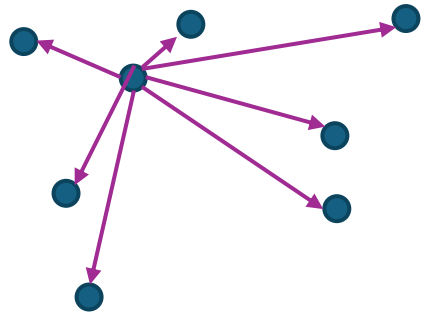
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  - Given the input (X, D), we start computing distances…
  - As long as we've only seen 1's, we cannot rule out a 2 still hiding out there
  - We must read all $\binom{n}{2}$ to know for sure.



Diameter Approximation

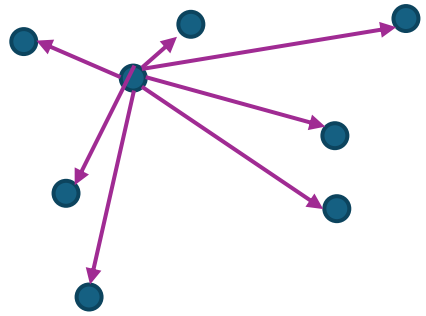
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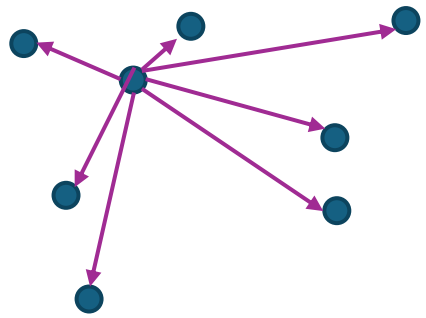
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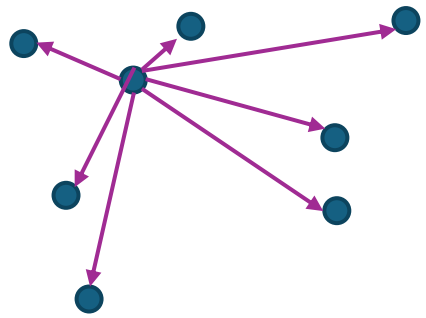
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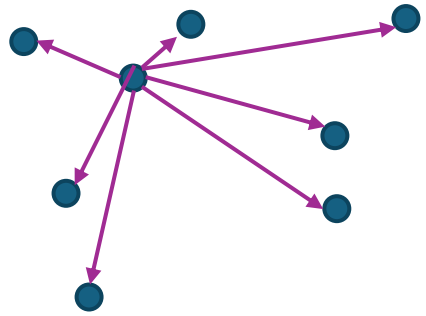
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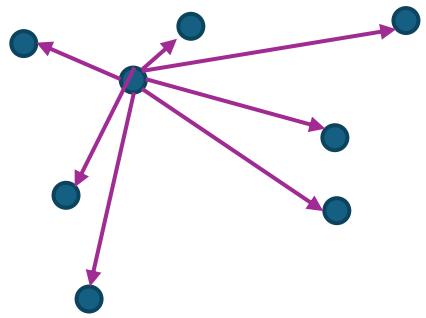
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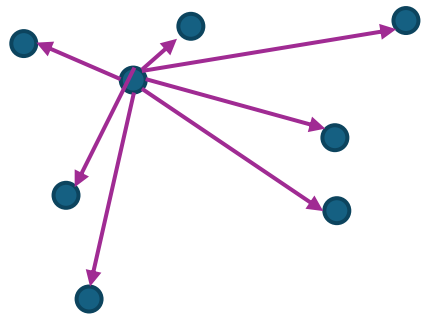
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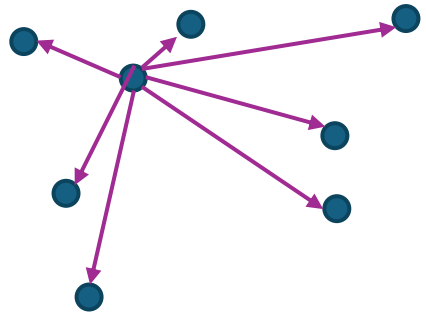
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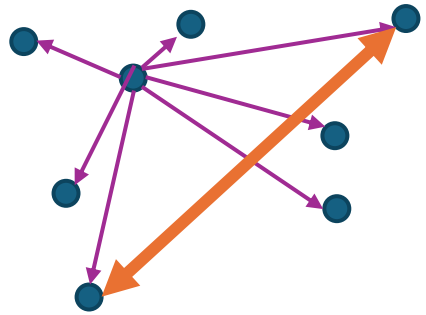
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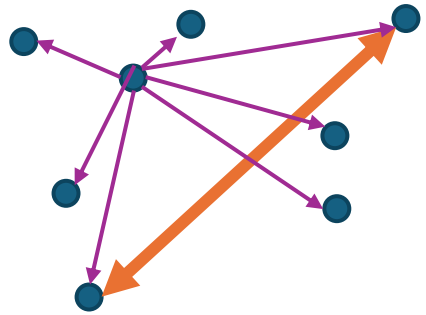
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Can we get a better approximation factor than $\frac{1}{2}$?



Diameter Lower Bound: Approximation

We proved:

- **Upper bound theorem:** $\widetilde{\Delta}$ satisfies: $\frac{1}{2}\mathrm{diam}_D(X) \leq \widetilde{\Delta} \leq \mathrm{diam}_D(X)$
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  - But what we *really* proved: Even if we are promised that the input is a (1,2)-metric, deciding if the diameter is 1 or 2 requires $\binom{n}{2} = \Omega(n^2)$ distance computations.
  - **Corollary**: For every $\alpha \in \left(0, \frac{1}{2}\right)$, computing a diameter approximation $\widetilde{\Delta}$ that satisfies $\alpha \cdot \mathrm{diam}_D(X) \leq \widetilde{\Delta} \leq \mathrm{diam}_D(X)$ requires $\binom{n}{2} = \Omega(n^2)$ distance computations.

Metric Embeddings

Metric Embeddings

- Definition: Let (X, D_X) and (Y, D_Y) be two metric spaces. A map $f: X \to Y$ is called an **isometric embedding** (or isometry) if for every $x, x' \in X$:

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- *Do you know any examples of metric embeddings?*

Isometric Embedding into ℓ_∞

- **Theorem:** Let (X, D) be a metric space with n points. Then it embeds isometrically into ℓ_∞ over $\mathbb{R}^n$ (with n dimensions).

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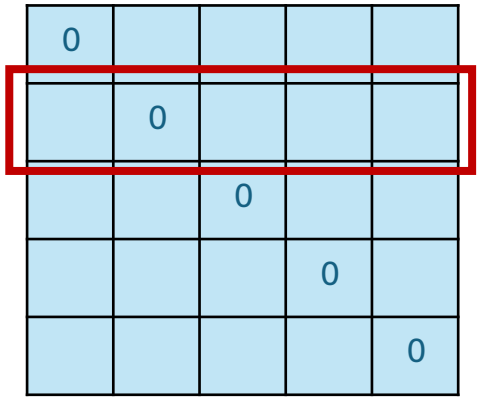
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Table of distances



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$$\|f(x_i) - f(x_k)\|_\infty = \max_{j=1,\dots,n} |f(x_i)_j - f(x_k)_j|$$

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0				
	0			
		0		
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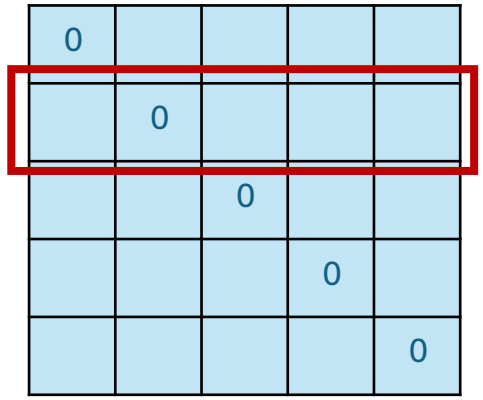
$$\|f(x_i) - f(x_k)\|_\infty = \max_{j=1,\dots,n} |f(x_i)_j - f(x_k)_j|$$

$$= \max_{j=1,\dots,n} |D(x_i, x_j) - D(x_k, x_j)| = D(x_i, x_k)$$

Why?

Table of distances

$f(x_i) =$



Isometric Embedding into ℓ_∞

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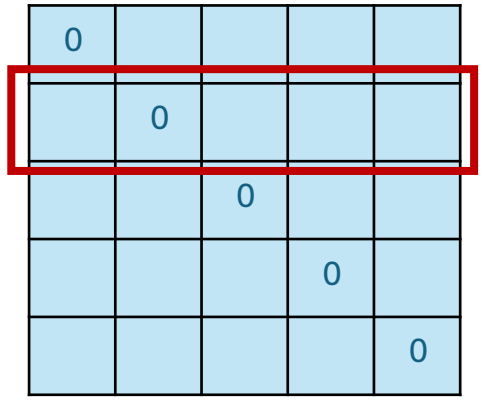
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Choose $j = i$ or $j = k$

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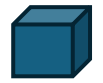
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	0			
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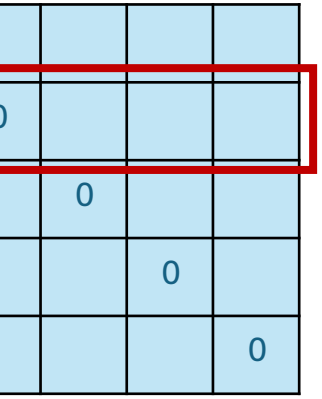


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How useful do you think this is algorithmically?

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Approximate Embedding

- Definition: Let (X, D_X) and (Y, D_Y) be two metric spaces. A map $f: X \to Y$ is called an **approximate metric embedding** if for every $x, x' \in X$:

$$\frac{1}{c} \cdot D_X(x, x') \leq D_Y\big(f(x), f(x')\big) \leq b \cdot D_X(x, x')$$

- for some $b, c \geq 1$. The approximation factor bc is called the **distortion**.
  - b is called the expansion factor
  - c is called the contraction factor

JL: Euclidean Dimension Reduction

- **Theorem** [Johnson-Lindenstrauss'84]: Let $n, d \in \mathbb{N}$ and $\varepsilon \in (0,1)$. For every $X \subset \mathbb{R}^d$ of size n, there exists a map $f: \mathbb{R}^d \to \mathbb{R}^m$ into dimension $m = O\left(\frac{\log n}{\varepsilon^2}\right)$, such that for every $x, y \in X$:

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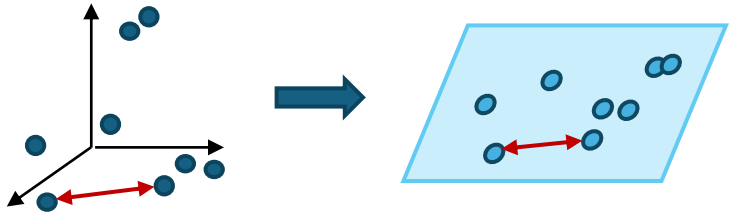
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- What do we need for this to be useful algorithmically?

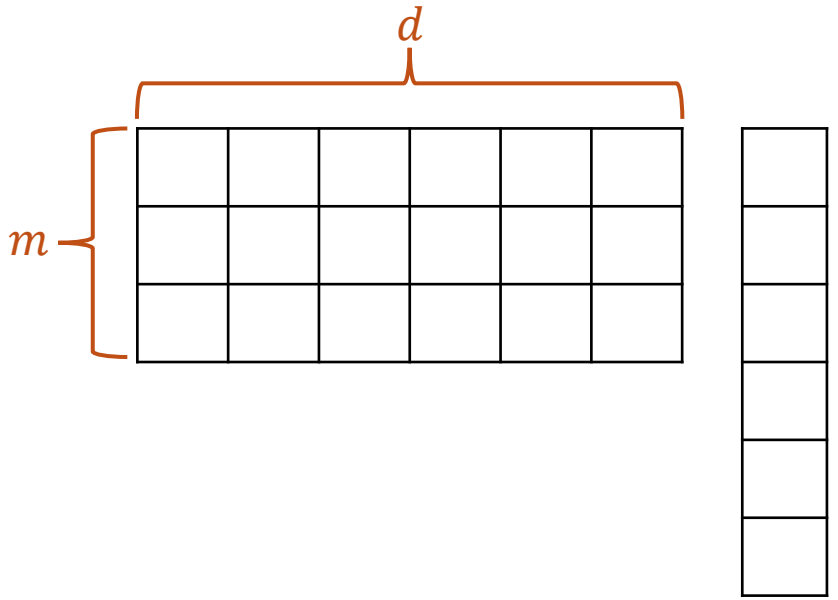
JL: Probabilistic Constructions

- There are many, for example (these two are essentially though not exactly the same):

Random projection:
Project all points onto a randomly chosen m-dimensional subspace



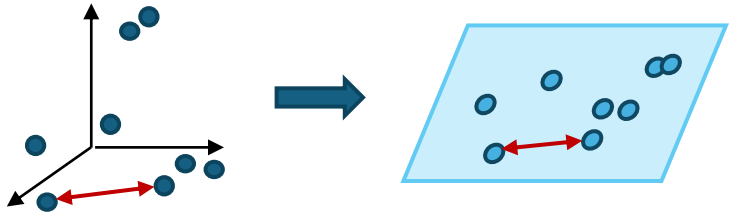
Random Gaussians:
Multiply all points by a $m \times d$ matrix with iid Gaussian entries



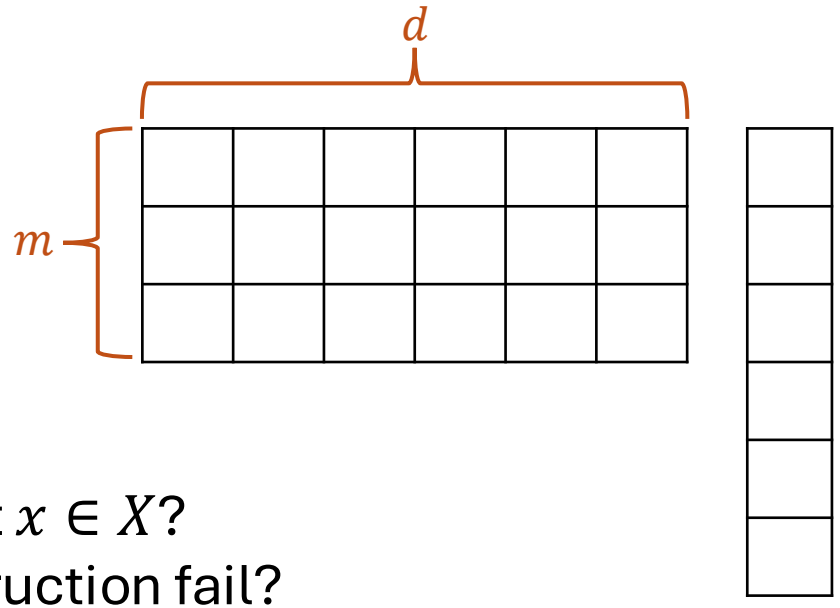
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Questions:

- What's the time to sample f?
- What's the time to apply f to a point $x \in X$?
- How often will a probabilistic construction fail?

JL via Random Gaussian Matrices

- **Theorem:** Let $n, d \in \mathbb{N}$ and $\varepsilon, \delta \in (0,1)$. There exists a distribution $\mathcal{F}_{n,d,\varepsilon,\delta}$ over maps $f: \mathbb{R}^d \to \mathbb{R}^m$ into dimension $m = O\left(\frac{\log(n/\delta)}{\varepsilon^2}\right)$, such that with probability $1 - \delta$, we have for every $x, y \in X$:

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- Note: This implies the JL theorem from before – "the probabilistic method"

Small Exercise



pe.app/talwagner

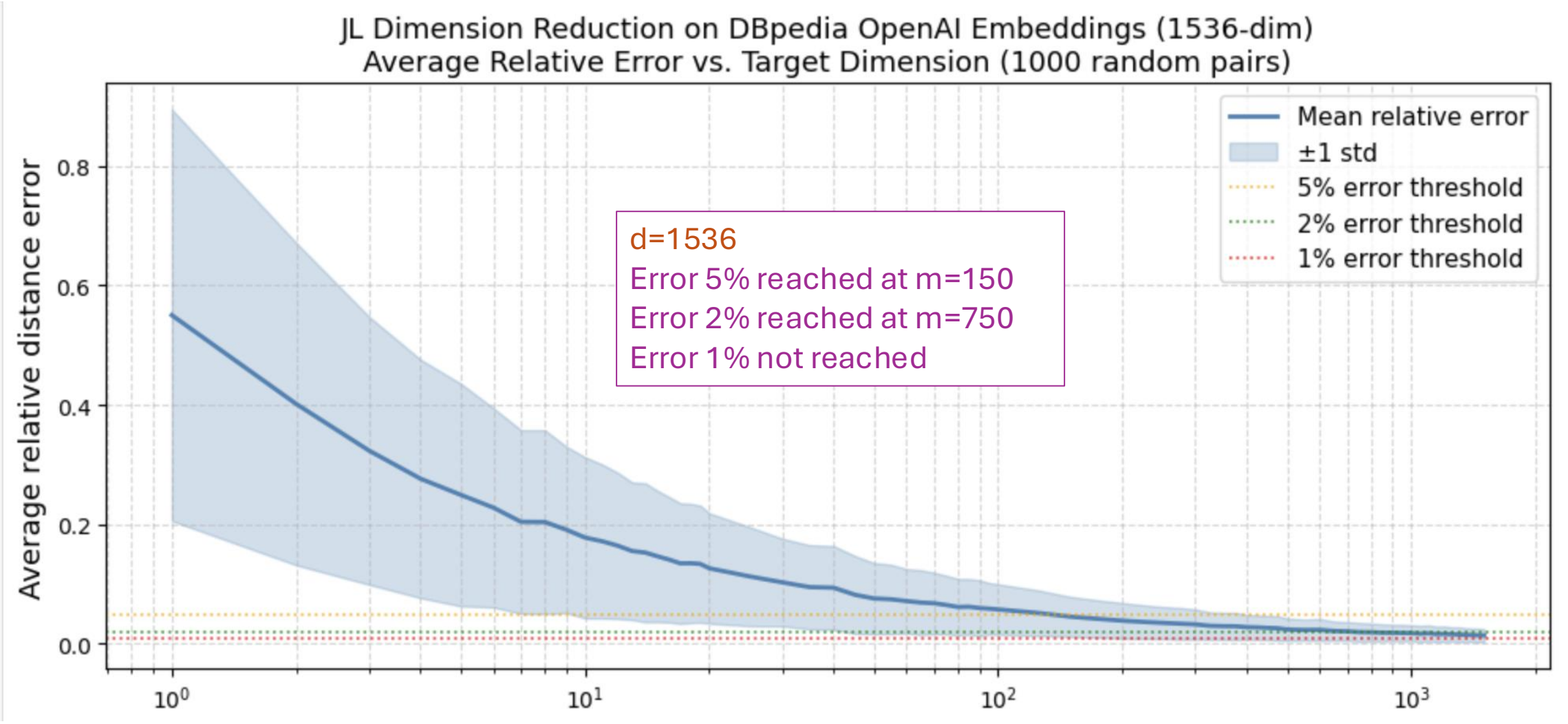
JL in Code

```
# n is the number of points
# d is the ambient (original) dimension
# m is the target dimension

# Sample a JL matrix
jl_matrix = np.random.standard_normal((d, m)) / np.sqrt(m)

# Project a dataset (given as an nxd matrix)
projected_dataset = dataset @ jl_matrix
```

JL in Action


JL Dimension Reduction on DBpedia OpenAI Embeddings (1536-dim)
Average Relative Error vs. Target Dimension (1000 random pairs)
Average relative distance error
0.0
0.2
0.4
0.6
0.8
10^0
10^1
10^2
10^3
Mean relative error
±1 std
5% error threshold
2% error threshold
1% error threshold
d=1536
Error 5% reached at m=150
Error 2% reached at m=750
Error 1% not reached

Proof of JL Theorem

- **Theorem:** Let $n, d \in \mathbb{N}$ and $\varepsilon, \delta \in (0,1)$. There exists $m = O\left(\frac{\log(n/\delta)}{\varepsilon^2}\right)$ such the the following holds. If $M \in \mathbb{R}^{m \times d}$ is a random matrix with iid entries from $N(0,1)$, then with probability $1 - \delta$, we have for every $x, y \in X$:

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- **We will now prove this**, through proving a single-vector version of the theorem, called the "distributional JL" (DJL) lemma. Plan:
  - **State DJL**
  - **Show that DJL ⇒ JL**
  - **Prove DJL**

Distributional JL (DJL): Focus on Single Vector

- **Lemma**: Let $\varepsilon, \delta \in (0,1)$ and $m = O\left(\frac{\log(1/\delta)}{\varepsilon^2}\right)$. Let $M \in \mathbb{R}^{m \times d}$ be a matrix with iid entries from $N(0,1)$. For every vector $z \in \mathbb{R}^d$ with unit norm $\|z\|_2 = 1$, we have

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- **Why does DJL imply JL?**
  - For every $x, y \in X$ we can apply the above lemma to the unit vector $z = \frac{x-y}{\|x-y\|_2}$:

$$1 - \varepsilon \leq \left\|\frac{1}{\sqrt{m}} Mz\right\|_2 \leq 1 + \varepsilon \iff (1-\varepsilon)\|x-y\|_2 \leq \left\|\frac{1}{\sqrt{m}} Mx - \frac{1}{\sqrt{m}} My\right\|_2 \leq (1+\varepsilon)\|x-y\|_2$$

  - We choose $\delta = \delta' / \binom{n}{2}$ and take a union bound over all pairs $x, y \in X$. Hence the $O\left(\frac{\log(n/\delta')}{\varepsilon^2}\right)$

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- **Chernoff-Hoeffding inequality**: If $X_1, \ldots, X_r$ are independent, and $X_i \in [a_i, b_i]$, and $X = \sum_i X_i$, then for every $t > 0$,

$$\Pr[|X - \mathbb{E}[X]| > t] \leq 2e^{-2t^2/\sum_{i=1}^{r}(b_i - a_i)^2}$$

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- **Proof** (for discrete random variables):
- $\mathbb{E}[X] = \sum_x x \cdot \Pr[X = x]$
- $\geq \sum_{x \geq a} x \cdot \Pr[X = x]$
- $\geq a \sum_{x \geq a} \Pr[X = x]$
- $= a \cdot \Pr[X \geq a]$

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- Proof:
  - Let $Y = (X - \mathbb{E}[X])^2$
  - Then $\mathbb{E}[Y] = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathrm{Var}(X)$
  - Then:
    - $\Pr[|X - \mathbb{E}[X]| \geq a] = \Pr[(X - \mathbb{E}[X])^2 \geq a^2]$
    - $= \Pr[Y \geq a^2]$
    - (Markov) $\leq a^{-2}\mathbb{E}[Y]$
    - $= a^{-2}\mathrm{Var}(X).$

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- Example: MGF of Bernoulli
  - Suppose $\Pr[X = 1] = p$ and $\Pr[X = 0] = 1 - p$
  - $\mathbb{E}\left[e^{\lambda X}\right] =$

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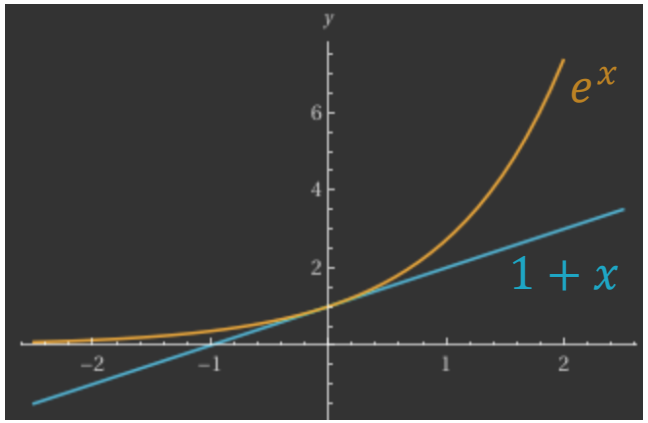
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  - Suppose $\Pr[X = 1] = p$ and $\Pr[X = 0] = 1 - p$
  - $\mathbb{E}\left[e^{\lambda X}\right] = p \cdot e^{\lambda} + 1 - p$

Probability Review: Moment Generating Function (MGF)

- The kth moment of a random variable X is $\mathbb{E}[X^k]$
- The moment generating function (MGF) of a random variable X is $M_X(\lambda) = \mathbb{E}\left[e^{\lambda X}\right]$
  - If the expectation exists in some interval $(-a, a)$

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  - Suppose $\Pr[X = 1] = p$ and $\Pr[X = 0] = 1 - p$
  - $\mathbb{E}\left[e^{\lambda X}\right] = p \cdot e^{\lambda} + 1 - p$
  - $= 1 + p\left(e^{\lambda} - 1\right)$
  - $\leq e^{p\left(e^{\lambda} - 1\right)}$



Chernoff Inequality: Proof

- **Chernoff inequality:** If $X_1, \ldots, X_r \in \{0,1\}$ are independent, and $X = \sum_i X_i$, then for every $\varepsilon \in [0,1]$, $\Pr\left[|X - \mathbb{E}[X]| \geq \varepsilon \mathbb{E}[X]\right] \leq 2e^{-\varepsilon^2 \mathbb{E}[X]/3}$.

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- Denote $p_i = \Pr[X_i = 1]$. Then $\mathbb{E}[X_i] = p_i$ and $\mathbb{E}[X] = \sum_i p_i$.

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  - It's left to show: $-\lambda(1+\varepsilon) + \left(e^{\lambda}-1\right) \leq -\frac{1}{3}\varepsilon^2$.
  - Choose $\lambda = \ln(1+\varepsilon)$ and use standard estimates for $\ln(1+\varepsilon)$.

Probability Review: Gaussians

- The Gaussian (or normal) distribution $N(\mu, \sigma^2)$ has pdf: $\varphi(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}$
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- Proof by MGF: $M_{X+Y}(\lambda) = \mathbb{E}\left[e^{\lambda(X+Y)}\right] = \mathbb{E}\left[e^{\lambda X}\right] \cdot \mathbb{E}\left[e^{\lambda Y}\right] = e^{\mu_1\lambda + 0.5\sigma_1^2\lambda^2} \cdot e^{\mu_2\lambda + 0.5\sigma_2^2\lambda^2}$

$$= e^{(\mu_1+\mu_2)\lambda + 0.5(\sigma_1^2+\sigma_2^2)\lambda^2}.$$



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- Consequence: Let $u \in \mathbb{R}^m$ be a random vector with iid entries from $N(0,1)$. Let $z \in \mathbb{R}^m$ be a fixed vector. Then,

$$u^T z = \sum_{i=1}^{d} z_i u_i \sim N\left(0, \sum_{i=1}^{d} z_i^2\right) = N(0, \|z\|_2^2) = \|z\|_2 N(0,1)$$

- Thus, $u^T z$ is distributed like $\|z\|_2 N(0,1)$.

(D)JL and 2-Stability

- **Recall DJL:**
- **Lemma**: Let $\varepsilon, \delta \in (0,1)$ and $m = O\left(\frac{\log(1/\delta)}{\varepsilon^2}\right)$. Let $M \in \mathbb{R}^{m \times d}$ be a matrix with iid entries from $N(0,1)$. For every vector $z \in \mathbb{R}^d$ with unit norm $\|z\|_2 = 1$, we have

$$\Pr\left[1 - \varepsilon \leq \left\|\frac{1}{\sqrt{m}} Mz\right\|_2 \leq 1 + \varepsilon\right] \geq 1 - \delta$$

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- Let $z \in \mathbb{R}^d$ be a given unit-norm vector
- Let $u \in \mathbb{R}^d$ be a single row of M (which is a vector with iid entries from $N(0,1)$)

$$\mathbb{E}[(u^T z)^2] = \mathbb{E}\left[\left(\|z\|_2 N(0,1)\right)^2\right] = \|z\|_2^2 \cdot \mathbb{E}\left[\left(N(0,1)\right)^2\right] = \|z\|_2^2$$

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- Therefore, denoting the rows of M by $u_1, \ldots, u_m$:

$$\mathbb{E}\left[\left\|\frac{1}{\sqrt{m}} Mz\right\|_2^2\right] = \mathbb{E}\left[\sum_{j=1}^{m} \left(\frac{1}{\sqrt{m}} u_j^T z\right)^2\right] = \frac{1}{m}\sum_{j=1}^{m} \mathbb{E}\left[\left(u_j^T z\right)^2\right] = \|z\|_2^2$$

Proof of DJL

$$\Pr\left[\left\|\frac{1}{\sqrt{m}}Mz\right\|_2^2 > 1+\varepsilon\right] = \Pr\left[\sum_{j=1}^{m}\left(u_j^T z\right)^2 > m(1+\varepsilon)\right]$$

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Denote: $\Psi = \sum_{j=1}^{m}\left(u_j^T z\right)^2$

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$$= e^{-\lambda m(1+\varepsilon)} \cdot \left(\frac{1}{\sqrt{1-2\lambda}}\right)^m$$

$$= e^{m(-\lambda\varepsilon-\lambda-0.5\ln(1-2\lambda))}$$

Plug the MGF of χ^2 (note: we need to make sure $\lambda < 0.5$)

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$$= e^{-\log(2/\delta)} = \frac{1}{2}\delta$$

$m = O\left(\frac{\log(1/\delta)}{\varepsilon^2}\right)$
Tune hidden constant in the $O(\cdot)$ according to c

Proof of DJL and JL: Conclusion

- We've proven the upper tail:

$$\Pr\left[\left\|\frac{1}{\sqrt{m}}Mz\right\|_2^2 > 1+\varepsilon\right] \leq \frac{1}{2}\delta$$

- The proof of the lower tail is very similar (exercise):

$$\Pr\left[\left\|\frac{1}{\sqrt{m}}Mz\right\|_2^2 < 1-\varepsilon\right] \leq \frac{1}{2}\delta$$

- By a union bound, we have proven DJL:

$$\Pr\left[1-\varepsilon \leq \left\|\frac{1}{\sqrt{m}}Mz\right\|_2^2 \leq 1+\varepsilon\right] \geq 1-\delta$$

- And, as shown earlier, DJL⇒JL theorem with random Gaussian matrices.



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- Can we make m smaller?
  - For **DJL**, $m = O\left(\frac{\log(1/\delta)}{\varepsilon^2}\right)$ is optimal [Jayram-Woodruff 2011, Kane-Meka-Nelson 2011]
  - For **JL**, things are a bit more subtle…

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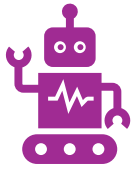
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  - **2026: AI proved the conjecture**



An Aside:

Algorithmic Sampling

How do we actually sample Gaussians (and other random variables)?

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How do we sample a random variable on a computer?

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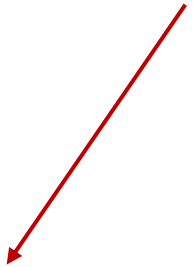
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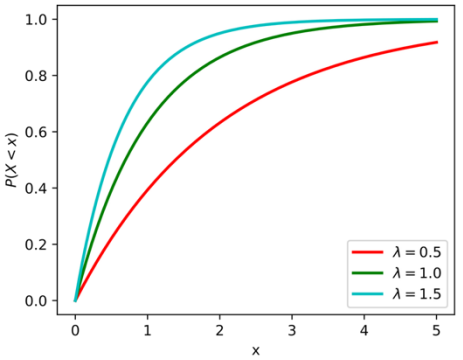
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- **Corollary**: If Φ_X is **invertible**, we can sample from X in time $O(1)$.
- Example: The exponential distribution $Exp(\lambda)$ has CDF $\Phi_{Exp(\lambda)}(x) = 1 - \mathrm{e}^{-\lambda x}$.
  Hence, we can sample $\hat{X} = Exp(\lambda)$ by $\hat{X} = -\lambda^{-1} \ln(1 - U)$

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- Option 2: Box-Muller transform
  1. Sample $U_1, U_2 \sim (0,1)$ iid
  2. Generate polar coordinates: $R = \sqrt{-2 \ln U_1}\,;\ \Theta = 2\pi U_2$
  3. Transfer to cartesian coordinates: $X_1 = R\cos\Theta\,;\ X_2 = R\sin\Theta$
  4. X_1, X_2 **are iid samples from** $N(0,1)$